

## PLANT PEST



appearance. Frank Peairs, Bugwood.org

### **Barley stripe rust (exotic strains)**

Exotic to Australia

Features: A fungal disease of barley that can cause complete crop losses

Where it's from: Asia, Europe, Central Africa and North, Central and South America

How it spreads: Importation of contaminated plant material, on machinery; spores on clothing and personal items

At risk: Barley

### **Keep it out**

This disease, caused by the fungus *Puccinia striiformis* f. sp. *hordei*, damages barley crops, sometimes completely wiping out crops.

The fungus produces stripes of rust pustules between the veins of leaves, and sometimes on barley heads. The pustules may be more yellow than orange, so the disease is sometimes referred to as yellow rust. Barley stripe rust can build up rapidly if conditions are cool and wet, ruining crops.

Rust spores are small, light and can survive away from plants for several days so that once introduced, the disease can be easily spread over large distances by wind and attached to clothing, machinery and tools.

Barley stripe rust poses a significant threat to Australia's barley crops since around 80 per cent of the varieties that we grow are susceptible.



Darrell Wesenberg, Bugwood.org



Jack Kelly Clark, UC Statewide IPM Program

## Importing goods

To keep barley stripe rust out of Australia, never ignore Australia's strict biosecurity rules.

Import shipments may need to be treated and certified, so before you import, check our [Biosecurity Import Conditions system \(BICON\)](#).

## What to look for

- Yellow patches within a barley crop.
- A close look will show stripe rust on leaves and barley heads.

Any barley plants showing stripe rust symptoms should be sent for identification through laboratory analysis.

## Where to look

### Importers

Importation of contaminated seed and straw is the most likely way that barley stripe rust would enter Australia.

### Grain growers

Check your barley crops frequently for the presence of new pests and diseases. Keep an eye out for any yellowing patches in the field.

If you visit places overseas that have the rust, be sure to clean clothes and boots before returning home.

## What to do

If you think you've found barley stripe rust in crops:

- take a photo
- do not disturb infected plants (this may be as simple as closing the doors on a shipping container or preventing access to a field).

Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/plant/identify/barley-stripe-rust>